

CPEN 416

Lecture 06: Multi-qubit measurements, the Bell basis, and superdense coding

Tuesday 29 September 2026

Announcements

- Midterm in class Tuesday 6 Oct (see PrairieLearn for details)
- Quiz 3 today
- Tutorial Assignment 1 due Friday 23:59

Last time

We used the tensor product to construct multi-qubit states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\varphi\rangle = \gamma|0\rangle + \delta|1\rangle,$$

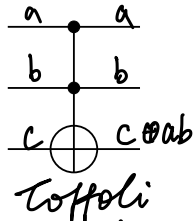
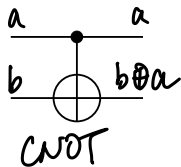
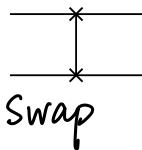
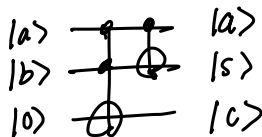
$$\{|0\rangle, |1\rangle\} \rightarrow \{|00\rangle, |01\rangle, \dots, |11\rangle\}$$

n qubit states $\rightarrow 2^n$ -dim- vector

$$\begin{aligned} |\psi\rangle \otimes |\varphi\rangle &= (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) \\ &= \alpha\gamma|00\rangle + \beta\gamma|10\rangle + \alpha\delta|01\rangle + \beta\delta|11\rangle \\ &= \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix} \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} \end{aligned}$$

Last time

We saw examples of two-qubit gates:



(reversible
AND)

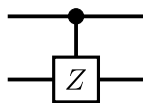
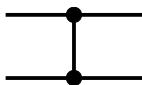
We used them to construct a small adder circuit.

Learning outcomes

- apply gates controlled on the state of multiple qubits
- define and give examples of entangled states
- measure multi-qubit systems in the computational basis and Bell basis
- leverage entanglement to implement superdense coding

Example: controlled-Z (CZ)

Exercise: what is the matrix representation of this operation?



$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = -|1\rangle$$

$$|00\rangle \rightarrow |00\rangle$$

$$|01\rangle \rightarrow |01\rangle$$

$$|10\rangle \rightarrow |10\rangle$$

$$|11\rangle \rightarrow -|11\rangle$$

$$Z = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

The bottom-right 2x2 submatrix is highlighted with a blue dashed box and labeled Z .

PennyLane: `qp.CZ`

$$CS = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & i \end{pmatrix}$$

The bottom-right 2x2 submatrix is highlighted with a blue dashed box and labeled S . A curved arrow labeled $e^{i\pi/2}$ points to the i element.

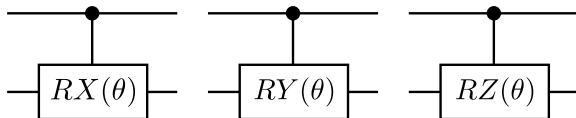
Image credit: Codebook node I.13

Example: controlled rotations (RX , RY , RZ)

We can control arbitrary Pauli rotations:

$$CRY(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ 0 & 0 & \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \quad RY(\theta)$$

Circuit elements:



PennyLane: `qp.CRX`, `qp.CRY`, `qp.CRZ`

Image credit: Codebook node I.14

Controlled- U

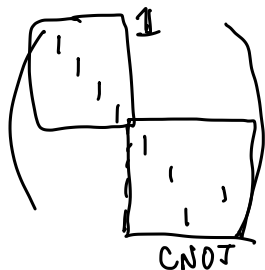
There is a pattern here:

$$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad CRY(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ 0 & 0 & \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$

More generally,

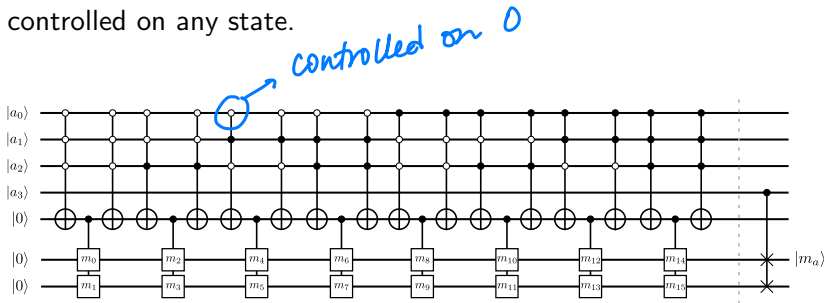
controlled U :

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & U \end{pmatrix}$$



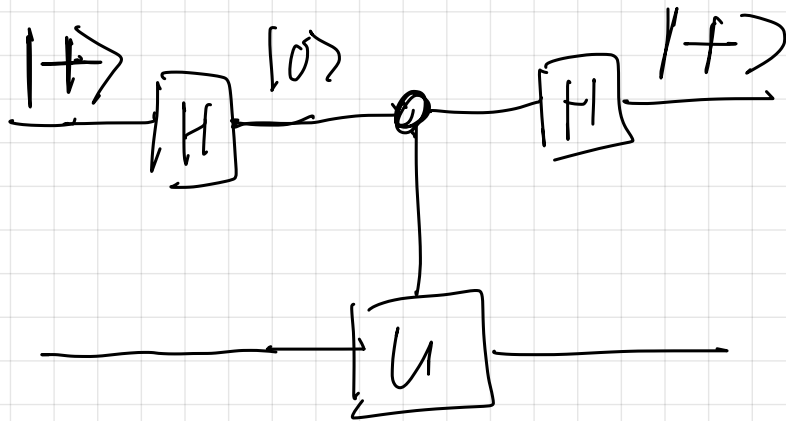
Controlled unitary operations

Any unitary operation can be turned into a controlled operation, controlled on any state.



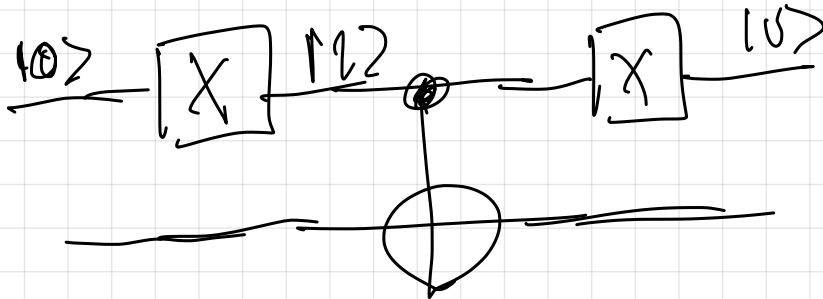
Most common controls are controlled-on- $|1\rangle$ (filled circle), and controlled-on- $|0\rangle$ (empty circle).

controlled on $|+\rangle$:



use basis rotation

controlled on $|0\rangle$??



Controlled unitary operations with `qp.ctrl`

Remember `qp.adjoint`:

```
@qp.qnode(dev)
def my_circuit():
    qp.S(wires=0)
    qp.adjoint(qp.S)(wires=0)
    return qp.sample()
```

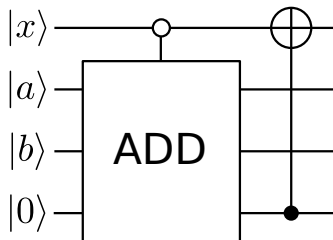
A similar *transform* allows us to perform arbitrary controlled operations (or entire quantum functions)!

```
@qp.qnode(dev)
def my_circuit():
    qp.S(wires=0)
    qp.ctrl(qp.S, control=1)(wires=0)
    return qp.sample()
```

Controlled unitary operations with `qp.ctrl`

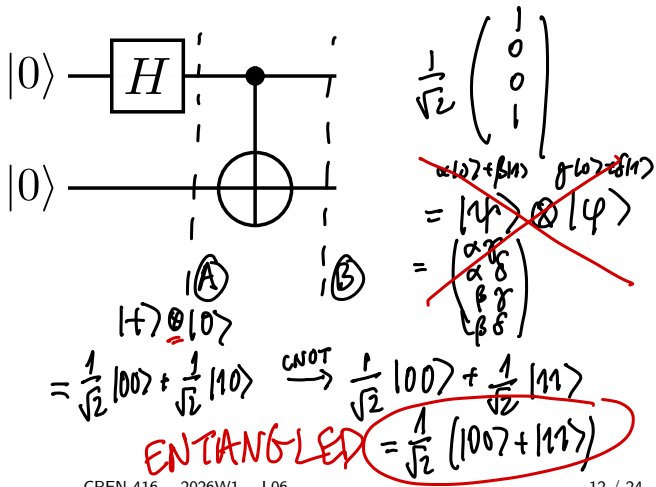
Take home exercise: using your adder from earlier, implement a circuit in PennyLane that uses `qp.ctrl` to

- perform addition if the control qubit is in state $|0\rangle$
- flip the state of the control qubit if the carry bit is 1



Controlling on superpositions

Exercise: Determine the combined output state of this circuit, then try to express it as a tensor product of two single-qubit states.



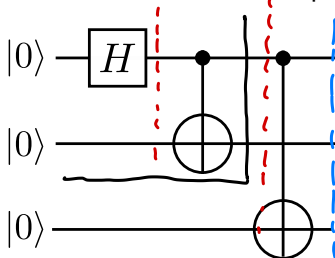
Entanglement

CNOT is an **entangling gate**!

*separable
1+78107*

Entanglement generalizes to more than two qubits:

*entangled (top two)
but separable
from bottom*



$\frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \otimes |0\rangle$

Exercise: Express the output state of this circuit in the computational basis.

*Greenberger, Horne
Zeilinger*

*$\frac{1}{\sqrt{2}} (|000\rangle + |111\rangle)$
GHZ state*

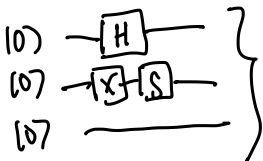
Multi-qubit measurement outcome probabilities

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

If we measure in the computational basis, the outcome probabilities are:

- $|\alpha|^2 = |\langle 00|\psi\rangle|^2$ for $|00\rangle$
- $|\beta|^2 = |\langle 01|\psi\rangle|^2$ for $|01\rangle$
- ...

Exercise: what is the probability of the qubits being in state $|110\rangle$ after measuring $(H \otimes SX \otimes I)|000\rangle$ in the computational basis?



output state:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes i|1\rangle \otimes |0\rangle$$

$$= \frac{1}{\sqrt{2}} i (|010\rangle + |110\rangle)$$

$\Rightarrow \text{prob} = \frac{1}{2}$

Multi-qubit measurement outcome probabilities

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

We can also measure *just one qubit*:

- The probability of the first qubit being in state $|0\rangle$ is $|\alpha|^2 + |\beta|^2$
- The probability of the second qubit being in state $|1\rangle$ is $|\beta|^2 + |\delta|^2$

Exercise: what is the probability of the second qubit being in state $|0\rangle$ after measuring $(H \otimes SX \otimes I)|000\rangle$ in the computational basis?

Next time

Bell basis measurements

Content: *Superdense coding*

- Quantum teleportation

Action items:

- 1 Tutorial assignment 1 due Friday 23:59
- 2 Study for midterm

Recommended reading:

- Codebook nodes IQC, SQ, MQ; Nielsen & Chuang 1.2, 1.3.1-1.3.7, 2.2.3-2.2.5, 2.3, 4.3